

ISAAC BERNAT

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Staff Software Engineer with 12+ years of experience in backend systems and product strategy. A proven leader of high-impact initiatives, including architecting a core subscription platform instrumental in a **Series C funding round** and delivering a **+3% global Gross Margin lift** on a separate monetization project.

PROFESSIONAL EXPERIENCE

- **AI Systems Engineering & R&D (Sabbatical)** Aug 2024 – Present
 - Architected Basepaint Media, a production-grade data pipeline for Generative AI. Built a custom Python orchestrator to manage LLM ingestion (Gemini), implementing **cost & context optimization, concurrency control, rate limiting** and **exponential backoff** using asyncio and tenacity. [🔗 Link: Project Repository](#)
 - Developed a CI/CD pipeline for a high-performance web portfolio, integrating automated accessibility testing and rigorous build processes to ensure engineering quality. [🔗 Link: Project Repository](#)

Preply

Mar 2020 – Jul 2024

Barcelona, Spain (Remote)

- **Staff Backend Engineer** Jun 2022 – Jul 2024
 - Drove the company's #1 most impactful initiative of 2023 (out of 215 A/B tests), delivering a **+3% global Gross Margin lift** by overhauling subscription cancellation and pause policies.
🔗 Case Study: Strategy & Data-Driven Decision Making
 - Acted as a strategic technical advisor, identifying critical risks in a major initiative and advocating for a phased approach that mitigated over **10,000 hours** in potential wasted development.
 - Proactively identified and mitigated critical financial exploits, including an automated abuse vector that leveraged payment processing thresholds. Designed and implemented a rate-limiting solution that eliminated the vulnerability while preserving legitimate user functionality.
 - Championed and improved engineering best practices, including advocating for enhanced observability and establishing a new, mandatory pre-launch monitoring policy that was adopted company-wide.
- **Senior Backend Engineer** Mar 2020 – Jun 2022
 - Served as the founding backend architect for the Subscription MVP, designed the core services and orchestrated the cross-team integrations (Payments, CRM, Data, Customer Success) required for a new recurring revenue system. This foundational work delivered a **+41% YoY increase in LTV**, improved the **LTV-to-CAC ratio from 1.4x to 3.0x** and was instrumental in securing a **Series C funding round**.
🔗 Case Study: MVP Architecture Product Strategy
 - Commanded 15+ production incidents and used the post-mortem from a critical SEV-2 to establish a mandatory, company-wide policy on pre-launch monitoring and observability.
🔗 Case Study: Incident Retrospective & Key Learnings
 - Led the backend migration of a high-traffic (1M+ MAU) application to a GraphQL API, improving page load times by **>85%** and unblocking A/B tests that resulted in an **+800% CVR increase**.

Ivbar Institute

Mar 2018 – Mar 2020

Stockholm, Sweden

- **Fullstack Engineer** Jan 2019 – Mar 2020
 - Engineered the UI for complex medical case-mix decision trees, empowering healthcare professionals to better analyze treatment pathways.
- **Backend Engineer** Mar 2018 – Jan 2019
 - Architected the core data processing engine for the Swedish National Quality Registry for Breast Cancer (NKBC), enabling nationwide analysis of treatment outcomes.
 - Implemented MECE-compliant privacy filters to ensure patient k-anonymity in highly sensitive medical datasets, a critical requirement for ethical data handling.

Productos Aditivos

Montcada, Spain

- **CTO & CFO** Jun 2016 – Jan 2018
 - Drove technology modernization and financial optimization, securing over **€64,000 in savings**.
 - Oversaw a critical ERP migration and the company-wide transition to the Spanish SII tax reporting system.

Wrapp

Oct 2012 – Jun 2016

Stockholm, Sweden

• Tech Lead

Sep 2014 – Jun 2016

- As a technical leader for the company's "2.0" pivot, guided the transition from a monolith to a distributed system of **50+ microservices**, owning 'users' (Python) and 'offers' (Go) critical backend services.
- Responsible for core backend APIs for mapping credit card transactions to merchant offers, the service architectural consistency and code quality across **1,200+ commits** from 12 contributors.
- Designed and built "Opsweb" from the ground up, a mission-critical internal deployment and CI/CD platform for managing canary releases of our 50+ microservices, directly improving developer velocity and site reliability.

• Early Engineer (Backend, Data, Frontend)

Oct 2012 – Sep 2014

- Developed and maintained a heuristic-based fraud prevention system to detect and block abuse of the platform's gift card and coupon features, protecting against financial loss from fake user profiles.
- Engineered foundational data systems, including the company's first content recommendation engine and a high-volume ETL pipeline for our Redshift data warehouse.
- Collaborated effectively in an asynchronous environments with teams in Stockholm and San Francisco.

FXstreet.com

Barcelona, Spain

• Software Developer

Feb 2012 – Jul 2012

- Contributed to a high-traffic C#/.NET financial news portal, handling complex internationalization (i18n) challenges across **17 languages**.

KEY PROJECTS & PRESENTATIONS

System Design: Heuristic-based Web Spam Classifier

- Authored a production-level technical design for a proof-of-concept spamdexing classifier, demonstrating a pragmatic approach to building new features within legacy ecosystems.

➔ [Link: Full Design Document](#)

PyCon Sweden 2019: Deep Dive on Python Performance

- Presented iterative code optimizations. Demonstrated how to diagnose, analyze and solve bottlenecks with CPython bytecode, profilers and low-level system tooling. Showcased techniques achieving **> 10¹⁸x speedup improvements**.

▶ [Link: Full Talk on YouTube](#)

PyCon Sweden 2014: Building Developer Tooling

- Selected to speak at the inaugural PyCon Sweden, presenting a talk on architecting and building 'pycrastinate', a language-agnostic tool for managing technical debt in large codebases using Git.

🔗 [Link: Project Repository](#)

Open Source: netflix-to-srt

- Creator and maintainer of a popular open-source tool with **★ 800+ on GitHub**. Solves a common community need for converting subtitle formats with emphasis on simplicity and portability.

🔗 [Link: Project Repository](#)

CORE COMPETENCIES

Programming Languages:

Python (Expert), Go, SQL, JavaScript, C++

Platforms & Databases:

AWS, PostgreSQL, Docker, Git, MySQL, Redshift, CI/CD (Jenkins, GitHub Actions)

Architecture & Design:

Distributed Systems, APIs (GraphQL, REST), AI-Assisted Development, Data Pipelines

Practices & Methodologies:

A/B Testing, TDD, Product Strategy, Incident Management, Cross-Functional Leadership

EDUCATION & LANGUAGES

M.Sc. in Informatics Engineering (EQF 7), Highest Honours

2006 – 2011

Universitat Politècnica de Catalunya (UPC - BarcelonaTech) & NTNU

Barcelona, Spain & Trondheim, Norway

- Master's Thesis on Test-Driven Conceptual Modelling conducted at the Norwegian University of Science and Technology (NTNU) as part of the ERASMUS programme.
- Awarded the highest possible grade (A with Highest Honours) from both institutions.

Languages

- **Native:** Catalan, Spanish • **Full Professional Command:** English (C2 Proficiency) • **Elementary:** Swedish